🏆 Infosys SE Elite Crack Vault — 06 Operating Systems and Complexity Elite

🎯 Infosys Systems Engineer (SE) | 📊 8 High-Yield Questions | 📅 5-6 Nov 2026 Drive | ⚡ 100% Pure SE Verified

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Problem 01 • Process States & Ready Queue

🔥 MUST-SOLVE

`ID: SEC03_EBOOK_Q004` `Source: 1. Pseudo Codes Technical ebook.pdf` `Level: Infosys SE Standard`

```
Where are placed the list of processes that are prepared to be executed and waiting for CPU allocation?
```

- a) Job queue
- b) Ready queue
- c) Execution queue
- d) Process queue

✓ **Verified Answer:** `Option B (Ready queue)`

Problem 02 • Process State Transitions & Blocking Primitives

🔥 MUST-SOLVE

`ID: SEC03_EBOOK_Q005` `Source: 1. Pseudo Codes Technical ebook.pdf` `Level: Infosys SE Standard`

```
Which of the following operations can block a running process?
```

- a) Fork
- b) Read
- c) Down (Semaphore P operation)
- d) All of these

✓ **Verified Answer:** `Option D (All of these)`

Problem 03 • Interrupts vs Operating System Software

MUST-SOLVE

ID: SEC03_EBOOK_Q006 | Source: 1. Pseudo Codes Technical ebook.pdf | Level: Infosys SE Standard

```
Which of the following does NOT interrupt a running process?
```

- a) Timer interrupt
- b) Device interrupt
- c) Power failure
- d) Scheduler process

✓ **Verified Answer:** `Option D (Scheduler process)`

Problem 04 • Matrix In-Place Double Swap Trap

MUST-SOLVE

ID: SEC03_EBOOK_Q025 | Source: 1. Pseudo Codes Technical ebook.pdf | Level: Infosys SE Standard

```
Let A be a square matrix of size n x n.
Consider the following program and state the output:

C = 100
for i = 1 to n do
    for j = 1 to n do
    {
        Temp = A[i][j] + C
        A[i][j] = A[j][i]
        A[j][i] = Temp - C
    }
for i = 1 to n do
    for j = 1 to n do
        Output(A[i][j]);
```

- a) The matrix A itself
- b) Transpose of matrix A
- c) Adding 100 to the upper diagonal
- d) None of the above

✓ **Verified Answer:** `Option A (The matrix A itself)`

Problem 05 • Asymptotic Time Complexity of Dependent Nested Loops

CORE SE

ID: SEC03_COMPLEX_01 | Source: CampusMonk Technical Pseudocode | Level: Infosys SE Standard

```
Integer a, b, n, count = 0
for (a = 1; a ⩽ n; a = a * 2)
{
    for (b = 1; b ⩽ a; b++)
    {
        count = count + 1
    }
}
Print count
```

What is the asymptotic time complexity of the above pseudocode fragment in terms of n?

- a) $O(n \log n)$
- b) $O(n)$
- c) $O(n^2)$
- d) $O(\log n)$

✓ **Verified Answer:** `Option B (O(n))`

Problem 06 • Four Coffman Conditions for Deadlock

CORE SE

ID: SEC03_OS_DEADLOCK_01 | Source: CampusMonk Operating Systems | Level: Infosys SE Standard

```
Which of the following is NOT one of the four necessary and sufficient Coffman conditions for a deadloc
```

- a) Mutual Exclusion
- b) Hold and Wait
- c) Preemption
- d) Circular Wait

✓ **Verified Answer:** `Option C (Preemption)`

Problem 07 • Round Robin CPU Scheduling & Quantum Selection

CORE SE

ID: SEC03_OS_RR_01 | Source: CampusMonk Operating Systems | Level: Infosys SE Standard

```
In Round Robin (RR) CPU scheduling, what happens if the time quantum (time slice) is set to an extremel
```

- a) The algorithm degenerates into Shortest Job First (SJF)
- b) The algorithm degenerates into First-Come First-Served (FCFS)
- c) The system enters a thrashing state
- d) Context switching overhead increases to infinity

✓ **Verified Answer:** `Option B (The algorithm degenerates into FCFS)`

Problem 08 • Virtual Memory & Page Replacement Thrashing

CORE SE

ID: SEC03_OS_VM_01 | Source: CampusMonk Operating Systems | Level: Infosys SE Standard

```
In virtual memory systems, what is the technical condition known as "Thrashing"?
```

- a) A condition where the CPU spends more time paging (swapping pages in and out) than executing user instructions
- b) A hardware deadlock occurring between the memory bus and the CPU L1 cache
- c) Fragmentation of the physical hard drive caused by unaligned sectors
- d) Premature termination of background daemon processes by the OOM killer

✓ **Verified Answer:** `Option A`